Portfolio Construction and Risk Analysis: Equal Weight, Minimum Variance, and Risk Parity

rayan nasloubi

March 2026

1 Introduction

Portfolio construction is a central problem in quantitative finance, as investors seek to balance expected returns against the level of risk they are willing to bear. The trade-off between risk and return has been extensively studied, starting from the classical mean-variance framework introduced by Markowitz (1952), which provides a formal approach to optimal asset allocation based on expected returns and the covariance structure of asset returns.

However, in practice, estimating expected returns is difficult and often leads to unstable portfolio allocations. As a result, alternative approaches that focus primarily on risk have gained popularity. Among these, the Minimum Variance portfolio aims to minimize overall portfolio volatility, while Risk Parity allocates capital such that each asset contributes equally to total portfolio risk. These methods are often considered more robust in empirical applications.

This project investigates whether more sophisticated allocation methods can improve the risk-return profile compared to a simple equally weighted portfolio. Using a diversified set of exchange-traded funds (ETFs) representing equities, bonds, and commodities, we construct and backtest three portfolio strategies: Equal Weight, Minimum Variance, and Risk Parity.

The objective is to evaluate and compare these strategies in terms of performance, volatility, and drawdown over a historical period. By doing so, this study aims to provide a clear and practical assessment of whether more advanced portfolio construction techniques deliver meaningful benefits in a realistic investment setting.

2 Data

The dataset used in this study consists of daily adjusted closing prices for a selection of exchange-traded funds (ETFs), chosen to represent a diversified set of asset classes. The selected assets are: SPY (U.S. large-cap equities), QQQ (U.S. technology equities), EFA (developed international equities), IEF (U.S. Treasury bonds), and GLD (gold).

These ETFs were selected to provide exposure to different sources of risk and return, including equities, fixed income, and commodities. This diversification is essential for evaluating portfolio allocation strategies such as Minimum Variance and Risk Parity, which rely on differences in volatility and correlation across assets.

The data is obtained using the `yfinance` Python library, which provides access to historical market data. The sample period spans from January 2015 to December 2025, and prices are collected at a daily frequency.

To ensure consistency, adjusted closing prices are used, as they account for corporate actions such as dividends and stock splits. Daily log returns are then computed from these prices and serve as the basis for all subsequent analysis, including the estimation of the covariance matrix and portfolio construction.

3 Methodology

3.1 Returns

Let P_t denote the adjusted closing price of an asset at time t. Daily log returns are computed as:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

Log returns are preferred due to their time-additive properties and their suitability for statistical modeling.

3.2 Covariance Matrix

Let $r = (r_1, r_2, \ldots, r_n)$ denote the vector of asset returns. The covariance matrix $\Sigma \in R^{n\times n}$ is defined as:

$$\Sigma_{ij} = \mathrm{Cov}(r_i, r_j)$$

This matrix captures the joint variability of asset returns and plays a central role in portfolio construction.

3.3 Portfolio Variance

Let $w = (w_1, w_2, \ldots, w_n)^T$ denote the vector of portfolio weights, with the constraint:

$$\sum_{i=1}^{n} w_i = 1$$

The variance of the portfolio is given by:

$$\sigma_p^2 = w^T \Sigma w$$

This quantity represents the total risk of the portfolio.

3.4 Equal Weight Portfolio

The Equal Weight portfolio assigns identical weights to each asset:

$$w_i = \frac{1}{n}, \quad \forall i$$

This strategy serves as a simple and robust benchmark.

3.5 Minimum Variance Portfolio

The Minimum Variance portfolio is obtained by solving the following optimization problem:

$$\min_{w} \; w^T \Sigma w$$

subject to:

$$\sum_{i=1}^{n} w_i = 1$$

and optionally:

$$w_i \geq 0 \quad \forall i$$

This approach aims to minimize overall portfolio volatility.

3.6 Risk Parity Portfolio

The Risk Parity approach seeks to allocate weights such that each asset contributes equally to total portfolio risk.

The marginal contribution to risk of asset i is given by:

$$\frac{\partial \sigma_p}{\partial w_i} = \frac{(\Sigma w)_i}{\sigma_p}$$

The risk contribution of asset i is:

$$RC_i = w_i \cdot \frac{(\Sigma w)_i}{\sigma_p}$$

Risk Parity imposes:

$$RC_i = RC_j \quad \forall i, j$$

This ensures that no single asset dominates the portfolio risk.

3.7 Backtesting Framework

To evaluate the performance of the different portfolio construction methods, a historical backtesting framework is implemented. The goal is to simulate the evolution of each portfolio over time under realistic rebalancing assumptions.

3.7.1 Estimation Window

At each rebalancing date, the covariance matrix Σ_t is estimated using a rolling window of past returns. More precisely, for each time t, the covariance matrix is computed using the previous T days of log returns:

$$\Sigma_t = \mathrm{Cov}(r_{t-T}, \ldots, r_{t-1})$$

In this study, a rolling window of $T = 252$ trading days (approximately one year) is used.

3.7.2 Rebalancing Strategy

Portfolios are rebalanced at a monthly frequency. At each rebalancing date:

- The covariance matrix Σ_t is estimated using past data.
- Portfolio weights w_t are computed according to each strategy (Equal Weight, Minimum Variance, Risk Parity).
- The portfolio is updated and held until the next rebalancing date.

This process ensures that portfolio allocations adapt dynamically to changing market conditions.

3.7.3 Portfolio Returns

The return of the portfolio at time t is computed as:

$$r_t^{(p)} = w_{t-1}^T r_t$$

where w_{t-1} are the portfolio weights determined at the previous rebalancing date.

The cumulative performance of the portfolio is then obtained by compounding returns over time:

$$V_t = V_0 \prod_{s=1}^{t} (1 + r_s^{(p)})$$

where V_0 is the initial portfolio value.

3.7.4 Performance Metrics

To compare the different strategies, several standard performance indicators are computed:

- Annualized return
- Annualized volatility
- Sharpe ratio

- Maximum drawdown

These metrics provide a comprehensive evaluation of both return and risk characteristics.

3.7.5 Visualization

The results are visualized using:

- Cumulative return curves
- Drawdown plots
- Time series of portfolio weights

These graphical representations allow for an intuitive comparison of the behavior of each strategy over time.

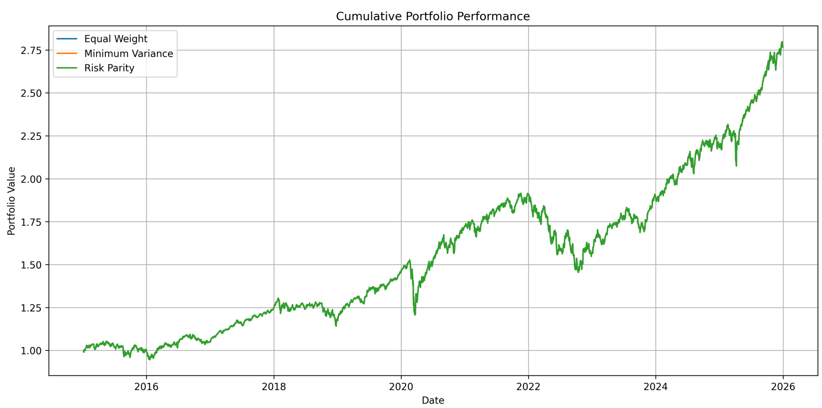

Figure 1: Cumulative performance of the Equal Weight, Minimum Variance, and Risk Parity portfolios.

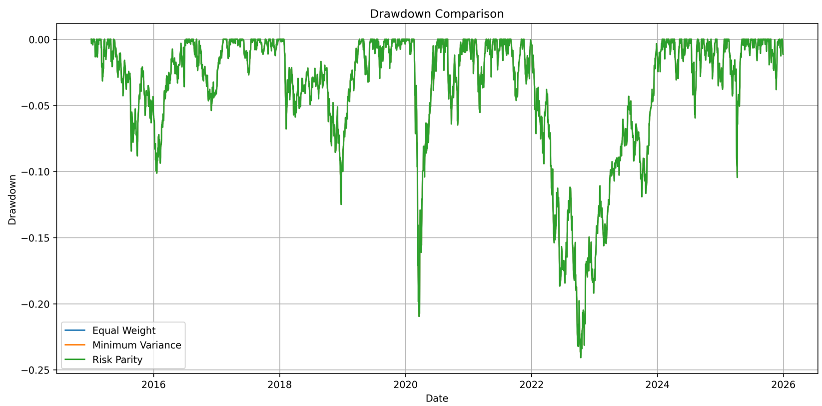

Figure 2: Drawdown comparison across the three portfolio strategies.

4 Conclusion

This project examined three portfolio construction approaches: Equal Weight, Minimum Variance, and Risk Parity, using a diversified set of exchange-traded funds. The objective was to assess whether more advanced allocation techniques could improve the trade-off between return and risk relative to a simple benchmark.

The results show that, in a static framework where the covariance matrix is estimated over the full sample period, all three methods converge toward similar portfolio allocations. This outcome highlights an important insight: when asset volatilities and correlations are relatively stable, optimization-based approaches may not provide a significant advantage over simple allocation rules such as Equal Weight.

However, this finding also underscores a key limitation of static portfolio construction. Financial markets are inherently dynamic, and the covariance structure of asset returns evolves over time. As a result, static optimization may fail to capture meaningful changes in risk conditions, leading to overly stable and potentially suboptimal allocations.

From a methodological perspective, this project illustrates the full pipeline of quantitative portfolio construction, including data collection, return computation, covariance estimation, optimization, and performance evaluation. It also emphasizes the sensitivity of portfolio outcomes to modeling assumptions, particularly the estimation window and the constraints imposed on portfolio weights.

Several limitations should be acknowledged. First, transaction costs are not included, which may materially affect the performance of rebalanced portfolios. Second, the analysis is conducted in-sample and does not include out-of-sample validation, limiting the ability to assess the robustness of the strategies. Third, the use of a simple covariance estimator may not adequately capture the true dynamics of financial markets.

Future work could extend this analysis by implementing rolling window estimation, allowing portfolio weights to adapt over time. Additional improvements could include the incorporation of transaction costs, the use of shrinkage estimators for covariance matrices, and the exploration of alternative allocation methods, including machine learning-based approaches.

Overall, this project provides a practical and rigorous introduction to quantitative portfolio construction and highlights both the strengths and the limitations of classical optimization techniques in real-world financial applications.